0.1 Product Space

Definition 0.1.1. Let $X_i,\ (i \in I)$ be Topological Spaces with Topology $\mathcal{T}_i$, respectively.
Define a *Product subbasis* on the Cartesian Product set $X = \prod_{i \in I} X_i$ as:

$$S_p \overset{\text{def}}{=} \{p_i^{-1}[U_i] \mid i \in I,\ U_i \in \mathcal{T}_i\}$$

where $p_i : X \to X_i : (x_i)_{i \in I} \mapsto x_i$.
The Topology $\mathcal{T}_p$ generated by S_p is called a *Product Topology* on X.

Clearly, the Product Topology is the smallest Topology such that:

all Projections are Continuous.

Moreover, on the Product Space, all Projections are Continuous Open Onto map.

Theorem 0.1.2. Let X and $Y_i,\ (i \in I)$ be topological spaces. Define a map

$$f : X \to \prod_{i \in I} Y_i : \quad x \mapsto (f_i(x))_{i \in I}$$

where each $f_i : X \to Y_i$ is a component function. Then,

f is continuous *if and only if* f_i is continuous for all $i \in I$

Proof. If f is continuous, then $f_i = p_i \circ f$ is continuous, being a composition of continuous maps.
Conversely, let $B \in \beta$ be a base element of $\mathcal{T}_P$. Then, by definition,

$$B = \bigcap_{i=1}^{n} p_i^{-1}[U_i] \text{ for some } U_i \in \mathcal{T}_i,\ i = 1, \dots, n.$$

Now,

$$f^{-1}[B] = f^{-1}\left[\bigcap_{i=1}^{n} p_i^{-1}[U_i]\right] = \bigcap_{i=1}^{n} (f^{-1} \circ p_i^{-1})[U_i] = \bigcap_{i=1}^{n} (p_i \circ f)^{-1}[U_i] = \bigcap_{i=1}^{n} f_i^{-1}[U_i].$$

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